

Main Tables

Paper: Olfactory dysfunction, daytime activity reduction, and 24-hour rhythm fragmentation — but not food-odor recognition — in NHANES 2013–2014

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Notes on formatting

- Cells show **β (lo, hi)** with significance markers: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
 - Coefficients on the native scale of each outcome.
 - M4 covariates: age, sex, race/ethnicity, education, PIR, BMI, smoking, diabetes, comorbidity count, PHQ-9, sinusitis, head injury, n medications.
 - Missing/non-estimable cells: ‘—’.
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Table 1. Sample characteristics by olfactory dysfunction status

Demographic, anthropometric, behavioral, and clinical characteristics of the analytic sample (n = 2,327), stratified by olfactory dysfunction (PST ≤ 5 correct). Continuous variables: mean (SD); categorical variables: %.

Characteristic	Overall	no-OD	OD	p
Age, years	57.7 (11.4)	57.0 (11.0)	64.5 (12.5)	<0.001
Female	52.4%	53.8%	39.5%	<0.001
Race/eth: NH White	73.7%	74.5%	66.2%	0.025
Race/eth: NH Black	10.4%	9.9%	15.0%	0.029
Race/eth: Hispanic	10.3%	10.1%	11.5%	0.892
Race/eth: NH Asian	3.8%	3.5%	6.5%	0.158
Education < HS	14.3%	13.5%	22.3%	<0.001
Education college+	32.4%	33.1%	26.5%	0.048
Poverty:income ratio	3.2 (1.6)	3.2 (1.6)	2.8 (1.5)	<0.001
BMI, kg/m ²	29.7 (7.0)	29.7 (6.9)	30.2 (8.0)	0.170
Current smoker	17.3%	17.8%	12.6%	0.102
Former smoker	29.3%	28.4%	38.2%	0.052
Diabetes (self-reported)	14.4%	13.7%	20.9%	0.050
HbA1c, %	5.8 (1.0)	5.8 (1.0)	5.9 (0.9)	0.281
Comorbidity count	1.0 (1.3)	1.0 (1.3)	1.2 (1.3)	0.007
PHQ-9 score	3.0 (4.2)	3.0 (4.2)	3.0 (3.9)	0.916
PHQ-9 ≥ 10	7.8%	7.9%	6.9%	0.929
PST score (0–8)	7.0 (1.2)	7.3 (0.8)	4.4 (1.0)	<0.001
Mean daily MIMS	13.8 (3.3)	14.0 (3.2)	12.6 (3.4)	<0.001
Sedentary, min/day	410.4 (89.8)	409.6 (89.7)	416.9 (90.3)	0.028
MVPA, min/day	28.9 (33.4)	30.0 (33.5)	19.2 (30.7)	<0.001
Total sleep time, min	342.0 (70.9)	342.9 (70.6)	332.9 (73.1)	0.047

Characteristic	Overall	no-OD	OD	p
Sleep efficiency	0.899 (0.048)	0.900 (0.047)	0.886 (0.052)	0.013
IS	0.592 (0.136)	0.593 (0.136)	0.579 (0.137)	0.111
IV	0.720 (0.226)	0.714 (0.217)	0.779 (0.288)	<0.001
RA	0.954 (0.039)	0.956 (0.036)	0.939 (0.061)	<0.001

Table 2. Aim 1 — Activity volume and intensity

*M1–M4 progressive adjustment of OD effects on mean MIMS, sedentary time, and MVPA. Cells: β (95% CI). Survey-weighted, cluster-robust regression with WTMEC2YR weights. $p < 0.05$, **$p < 0.01$** , $p < 0.001$. $n = 2,055$ (M4).*

Outcome	M1 (Unadjusted)	M2 (+ demographics)	M3 (+ comorbidities)	M4 (+ depression, ENT, meds)
Mean daily MIMS	-1.399 (-1.904, -0.895)***	-0.738 (-1.189, -0.288)**	-0.652 (-0.970, -0.334)***	-0.612 (-0.930, -0.294)***
Sedentary (min/d)	7.300 (-5.800, 20.40)	0 (-13.10, 13.20)	-1.900 (-14.60, 10.80)	-2.900 (-16.60, 10.70)
MVPA (min/d)	-10.90 (-14.50, -7.200)***	-4.500 (-7.600, -1.300)**	-3.800 (-6.600, -1.000)*	-3.400 (-6.400, -0.400)*

Table 3. Aims 2 & 3 — 24-hour rhythm, sleep, and fragmentation

*M1–M4 progressive adjustment of OD effects on rhythm metrics (IS, IV, RA), sleep parameters (TST, WASO, sleep efficiency), and per-person ASTP. Cells: β (95% CI). $p < 0.05$, **$p < 0.01$** , $p < 0.001$. $n = 2,055$ or 1,683 (M4).*

Outcome	M1 (Unadjusted)	M2 (+ demographics)	M3 (+ comorbidities)	M4 (+ depression, ENT, meds)
Interdaily stability (IS)	-0.014 (-0.037, 0.0080)	-0.031 (-0.059, -0.0040)*	-0.030 (-0.055, -0.0050)*	-0.028 (-0.053, -0.0040)*
Intradaily variability (IV)	0.065 (0.027, 0.103)**	0.053 (0.014, 0.092)**	0.051 (0.014, 0.088)**	0.048 (0.011, 0.084)*
Relative amplitude (RA)	-0.016 (-0.025, -0.0080)***	-0.0080 (-0.017, 0.0010)	-0.0070 (-0.016, 0.0010)	-0.0070 (-0.015, 0.0020)
Total sleep time (min)	-10.00 (-19.30, -0.600)*	-3.500 (-14.60, 7.500)	-3.700 (-13.40, 6.000)	-2.800 (-12.70, 7.200)
Sleep efficiency	-0.015 (-0.023, -0.0060)**	-0.0070 (-0.018, 0.0040)	-0.0070 (-0.017, 0.0040)	-0.0060 (-0.015, 0.0040)
WASO (min)	4.800 (1.400, 8.100)**	2.700 (-0.900, 6.200)	2.300 (-1.200, 5.800)	2.200 (-1.200, 5.600)
ASTP	0.021 (0.0060, 0.035)**	0.014 (0.0030, 0.025)*	0.012 (0.0040, 0.021)**	0.011 (0.0030, 0.020)**

Table 4. Effect modification by age, sex, smoking, and depression

*OD × modifier interactions across five primary outcomes. M4-adjusted, stratified estimates. Cells in last two columns: β (95% CI). Significance markers on interaction p: * $p < 0.05$, ** $p < 0.01$. $n = 2,055$.*

Outcome	Modifier	Interaction p	Effect at modifier=0	Effect at modifier=1
Mean MIMS/day	Age ≥60 (vs 40-59)	0.010**	0.071 (-0.675 to 0.817)	-1.336 (-1.878 to -0.793)
Mean MIMS/day	Female (vs male)	0.543	-0.699 (-1.094 to -0.303)	-0.483 (-1.047 to 0.081)
Mean MIMS/day	Ever smoker (vs never)	0.170	-0.867 (-1.350 to -0.385)	-0.335 (-0.881 to 0.211)
Mean MIMS/day	PHQ-9 ≥10 (vs <10)	0.094	-0.554 (-0.907 to -0.201)	-1.388 (-2.259 to -0.517)
IV	Age ≥60 (vs 40-59)	0.020*	-0.031 (-0.110 to 0.048)	0.108 (0.048 to 0.167)
IV	Female (vs male)	0.226	0.027 (-0.023 to 0.078)	0.078 (0.017 to 0.138)
IV	Ever smoker (vs never)	0.473	0.029 (-0.046 to 0.105)	0.065 (0.018 to 0.113)
IV	PHQ-9 ≥10 (vs <10)	0.919	0.047 (0.009 to 0.086)	0.054 (-0.079 to 0.187)
ASTP	Age ≥60 (vs 40-59)	0.035*	-0.005 (-0.024 to 0.013)	0.024 (0.010 to 0.038)
ASTP	Female (vs male)	0.108	0.017 (0.006 to 0.028)	0.003 (-0.010 to 0.015)
ASTP	Ever smoker (vs never)	0.306	0.016 (0.008 to 0.025)	0.006 (-0.011 to 0.023)
ASTP	PHQ-9 ≥10 (vs <10)	0.086	0.010 (0.000 to 0.019)	0.033 (0.010 to 0.056)